Time Series Analysis & Classification

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Chapter V: Model Specification

1. The sample autocorrelation function
2. Partial Autocorrelation Function (PACF)
3. Assessing Nonstationarity
4. Other model selection methods
5. Summary

Introduction

- In this chapter, we discuss techniques on how to choose suitable values of p, d, and q for an observed (or transformed) time series.
- We want our choices to be consistent with the underlying structure of the observed data.
- Bad choices of p, d, and q lead to bad models, which, in turn, lead to bad predictions (forecasts) of future values.

The sample autocorrelation function

The sample autocorrelation function

Recall: For time series data $Y_1, Y_2, \ldots, Y_n$, the sample autocorrelation function (ACF), at lag k, is given by

$$r_k = \frac{\sum_{t=k+1}^{n}(Y_t - \bar{Y})(Y_{t-k} - \bar{Y})}{\sum_{t=1}^{n}(Y_t - \bar{Y})^2},$$

where $\bar{Y}$ is the sample mean of $Y_1, Y_2, \ldots, Y_n$.

The sample autocorrelation function

- The sample autocorrelation r_k is an estimate of the true (population) autocorrelation ρ_k. As with any statistic, r_k has a sampling distribution which describes how it varies from sample to sample.

- We would like to know this distribution so we can quantify the uncertainty in values of r_k that we might see in practice.

The sample autocorrelation function

Theory: For a stationary $ARMA(p, q)$ process,

$$\sqrt{n}(r_k - \rho_k) \xrightarrow{d} \mathcal{N}(0, c_{kk}),$$

as $n \to \infty$, where

$$c_{kk} = \sum_{l=-\infty}^{\infty} (\rho_l^2 + \rho_{l-k}\rho_{l+k} - 4\rho_k\rho_l\rho_{l-k} + 2\rho_k^2\rho_l^2).$$

In other words, when the sample size n is large, the sample autocorrelation r_k is approximately normally distributed with mean ρ_k and variance c_{kk}/n:

$$r_k \sim \mathcal{N}(\rho_k, c_{kk}/n).$$

The sample ACF: White Noise

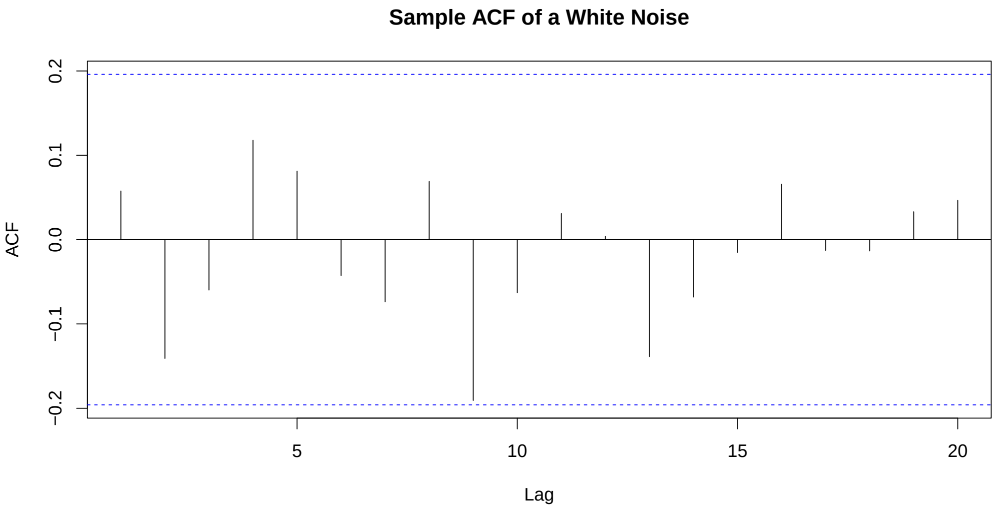
- For a white noise process, the formula for c_{kk} simplifies considerably because nearly all the terms in the sum above are zero. For large n,

$$r_k \sim \mathcal{N}\left(0, \frac{1}{n}\right),$$

  for $k = 1, 2, \ldots$.

- Thus, we have $\pm 2/\sqrt{n}$ as approximate margin of error bounds for r_k. Values of r_k outside these bounds would be "unusual" under the white noise model assumption.

The sample ACF: White Noise



The sample ACF: $MA(q)$

- For an invertible $MA(q)$ process, and
  for $k > q$, the sample autocorrelation r_k satisfies (when n is large):

$$r_k \sim \mathcal{N}\left[0, \frac{1}{n}\left(1 + 2\sum_{j=1}^{q} \rho_j^2\right)\right],$$

- This suggests a large-sample test for:

$$H_0 : MA(q) \text{ process is appropriate} \quad \text{vs} \quad H_1 : MA(q) \text{ process is not appropriate}$$

The sample ACF: $MA(q)$

- Under H_0, $r_{q+1} \sim \mathcal{N}\left[0, \frac{1}{n}\left(1 + 2\sum_{j=1}^{q} \rho_j^2\right)\right]$.

- Define Z^* (where r_j are estimates of ρ_j):

$$Z^* = \frac{r_{q+1}}{\sqrt{\frac{1}{n}\left(1 + 2\sum_{j=1}^{q} \rho_j^2\right)}} \sim \mathcal{N}(0, 1). \qquad (\text{Under } \; H_0)$$

- Reject H_0 if $|Z^*| > z_{\alpha/2}$, where $z_{\alpha/2}$ is the upper $\alpha/2$ quantile of the standard normal distribution. (Two-sided test). Or, equivalently, reject H_0 if the (two-sided) p-value is less than α.

Partial Autocorrelation Function (PACF)

Partial Autocorrelation Function (PACF)

Recall:

- In $MA(q)$ models, ACF $\rho_k \neq 0$ for $k \leq q$ and $\rho_k = 0$ for larger lags.
- ACF helps identify dependence order in MA models.
- ACF may not reveal dependence order in AR models.

We introduce the PACF for AR models, similar to ACF for MA models.

Partial Autocorrelation Function (PACF)

Motivation ($AR(1)$ example):

- Consider $Y_t = \phi Y_{t-1} + e_t$, where e_t is white noise.
- If Y_t followed $MA(1)$, then $\text{cov}(Y_t, Y_{t-2}) = 0$.
- For $AR(1)$, $\text{Cov}(Y_t, Y_{t-2}) \neq 0$ due to dependence through Y_{t-1}.

Strategy

- Suppose that we break the dependence between Y_t and Y_{t-2} in an $AR(1)$ process by removing (or partialing out) the effect of Y_{t-1}.
- To do this, consider the quantities $Y_t - \phi Y_{t-1}$ and $Y_{t-2} - \phi Y_{t-1}$. Note that:

$$\begin{aligned} \mathrm{Cov}(Y_t - \phi Y_{t-1}, Y_{t-2} - \phi Y_{t-1}) &= \mathrm{Cov}(e_t, Y_{t-2} - \phi Y_{t-1}) \\ &= 0, \end{aligned}$$

  because e_t is independent of Y_{t-1} and Y_{t-2}.

Strategy

Now, we make the following observations.

- In the $AR(1)$ model, if ϕ is known, we can think of $Y_t - \phi Y_{t-1}$ as the prediction error from regressing Y_t on Y_{t-1} (with no intercept; this is not needed because we are assuming a zero mean process).
- Similarly, the quantity $Y_{t-2} - \phi Y_{t-1}$ can be thought of as the prediction error from regressing Y_{t-2} on Y_{t-1}, again with no intercept.
- Both of these prediction errors are uncorrelated with the intervening variable Y_{t-1}.

Strategy

- To see why, note that

$$\begin{aligned}\mathrm{Cov}(Y_t - \phi Y_{t-1}, Y_{t-1}) &= \mathrm{Cov}(Y_t, Y_{t-1}) - \phi\,\mathrm{Cov}(Y_{t-1}, Y_{t-1}) \\ &= \gamma_1 - \phi\gamma_0 = 0,\end{aligned}$$

  because $\gamma_1 = \phi\gamma_0$ in the $AR(1)$ model.

- An identical argument shows that

$$\mathrm{Cov}(Y_{t-2} - \phi Y_{t-1}, Y_{t-1}) = \gamma_1 - \phi\gamma_0 = 0.$$

Terminology

- For a zero-mean time series, let $\hat{Y}_t^{(k-1)}$ denote the population regression of Y_t on the variables $Y_{t-1}, Y_{t-2}, \ldots, Y_{t-(k-1)}$, that is,

$$\hat{Y}_t^{(k-1)} = \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \cdots + \beta_{k-1} Y_{t-(k-1)}.$$

- Let $\hat{Y}_{t-k}^{(k-1)}$ denote the population regression of Y_{t-k} on the variables $Y_{t-1}, Y_{t-2}, \ldots, Y_{t-(k-1)}$, that is,

$$\hat{Y}_{t-k}^{(k-1)} = \beta_1 Y_{t-(k-1)} + \beta_2 Y_{t-(k-2)} + \cdots + \beta_{k-1} Y_{t-1}.$$

- The partial autocorrelation function (PACF) of a stationary process $\{Y_t\}$, denoted by ϕ_{kk}, satisfies $\phi_{11} = \rho_1$ and

$$\phi_{kk} = \operatorname{corr}\left(Y_t - \hat{Y}_t^{(k-1)}, Y_{t-k} - \hat{Y}_{t-k}^{(k-1)}\right), \quad \text{for } k = 2, 3, \ldots.$$

Terminology

- If the underlying process $\{Y_t\}$ is normal, then an equivalent definition is

$$\phi_{kk} = \operatorname{corr}(Y_t, Y_{t-k} | Y_{t-1}, Y_{t-2}, \ldots, Y_{t-(k-1)}),$$

  the correlation between Y_t and Y_{t-k}, conditional on the intervening variables $Y_{t-1}, Y_{t-2}, \ldots, Y_{t-(k-1)}$.

- That is, ϕ_{kk} measures the correlation between Y_t and Y_{t-k} after removing the linear effects of $Y_{t-1}, Y_{t-2}, \ldots, Y_{t-(k-1)}$.

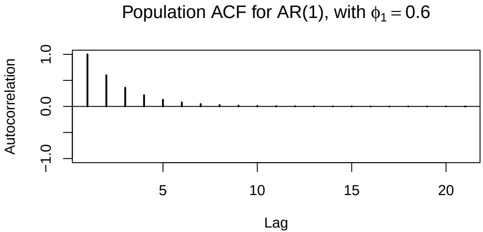
General Result for $AR(p)$ Processes

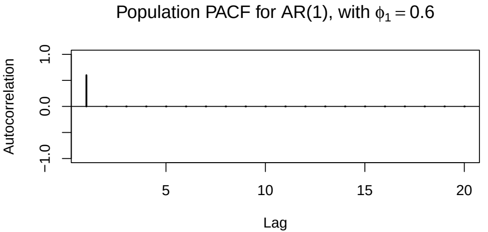
For an $AR(p)$ process, the first p partial autocorrelations are nonzero, while the remaining ones are zero.

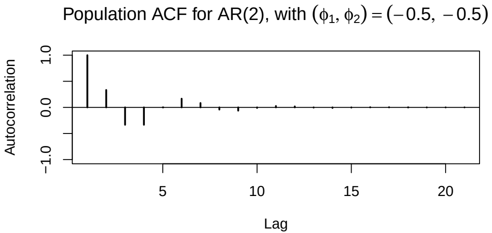
- $\phi_{11} = \phi_{22} = \cdots = \phi_{pp} \neq 0$; i.e., the first p partial autocorrelations are nonzero.
- $\phi_{kk} = 0$, for all $k > p$.

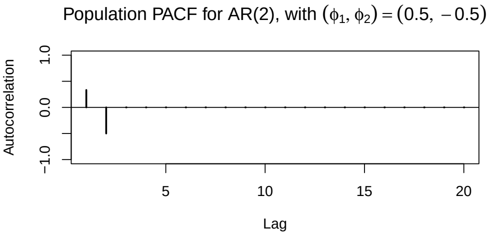
This property makes the PACF a useful tool for identifying the order of an $AR(p)$ model.

General Result for $AR(p)$ Processes









Result

- When the $AR(p)$ model is correct, then for large n, and for all $k > p$:

$$\hat{\phi}_{kk} \sim \mathcal{AN}\left(0, \frac{1}{n}\right),$$

- Therefore, we can use $\pm z_{\alpha/2}/\sqrt{n}$ as critical points to test, at level α:

$$H_0 : AR(p) \text{ model is appropriate} \quad \text{vs} \quad H_1 : AR(p) \text{ model is not appropriate},$$

in the same way that we tested whether or not a specific MA model was appropriate using the sample autocorrelations r_k.

Comparison with ACF

- The PACF can help determine the order of an $AR(p)$ process just like the ACF can help determine the order of an $MA(q)$ process!

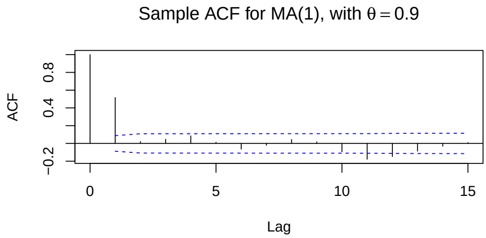
Comparison

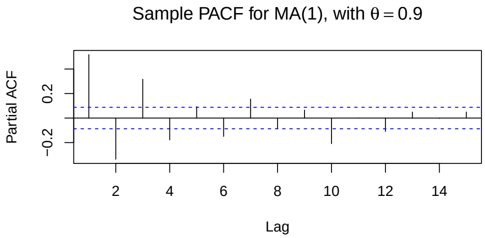
- The following table summarizes the behavior of the ACF and PACF for moving average and autoregressive processes.

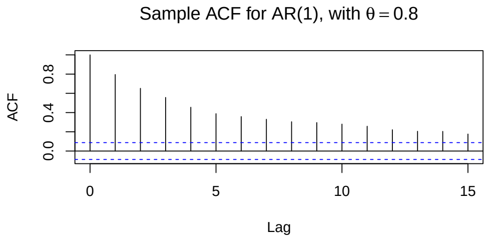
	$AR(p)$	$MA(q)$
ACF	Tails off	Cuts off after lag q
PACF	Cuts off after lag p	Tails off

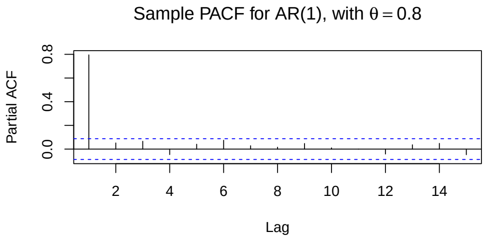
- Therefore, the ACF is the key tool to help determine the order of a MA process. The PACF is the key tool to help determine the order of an AR process.

Examples

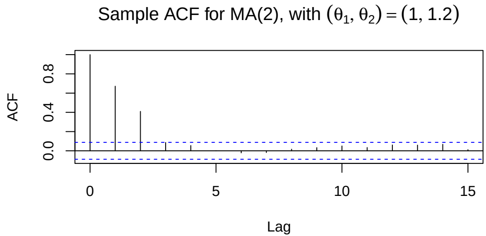

Sample ACF for MA(1), with θ = 0.9
ACF
−0.2
0.4
0.8
0
5
10
15
Lag

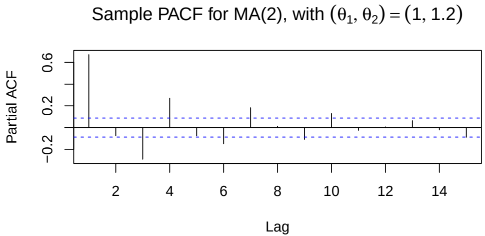

Sample PACF for MA(1), with θ = 0.9
Partial ACF
−0.2
0.2
2
4
6
8
10
12
14
Lag

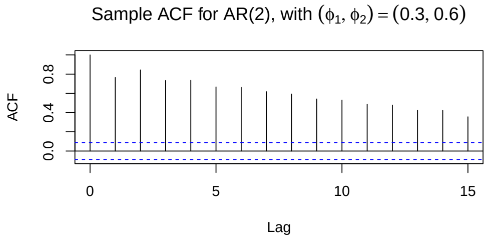

Sample ACF for AR(1), with θ = 0.8
ACF
0.0
0.4
0.8
0
5
10
15
Lag

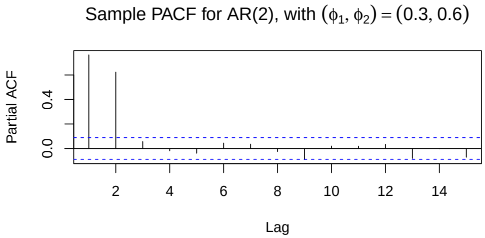

Sample PACF for AR(1), with θ = 0.8
Partial ACF
0.0
0.4
0.8
2
4
6
8
10
12
14
Lag

Examples

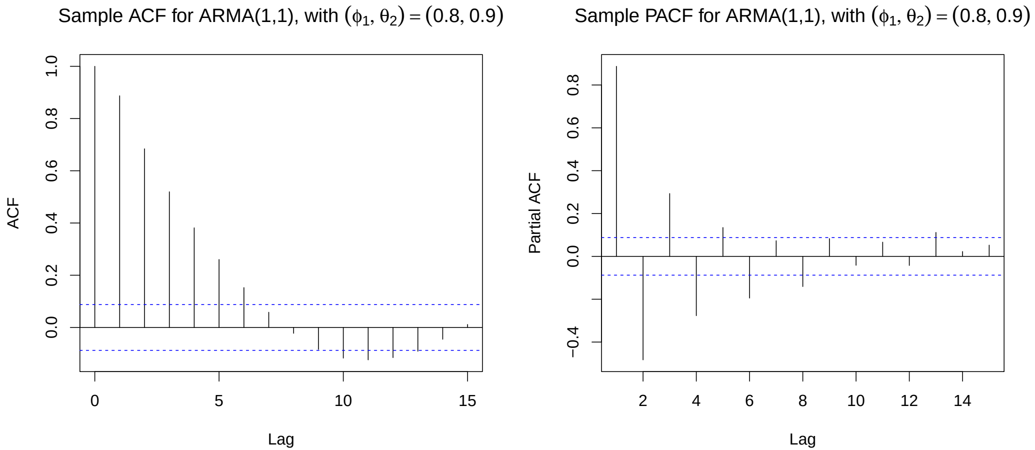

Sample ACF for MA(2), with $(\theta_1, \theta_2) = (1, 1.2)$
ACF
0.0
0.4
0.8
0
5
10
15
Lag


Sample PACF for MA(2), with $(\theta_1, \theta_2) = (1, 1.2)$
Partial ACF
−0.2
0.2
0.6
2
4
6
8
10
12
14
Lag


Sample ACF for AR(2), with $(\phi_1, \phi_2) = (0.3, 0.6)$
ACF
0.0
0.4
0.8
0
5
10
15
Lag


Sample PACF for AR(2), with $(\phi_1, \phi_2) = (0.3, 0.6)$
Partial ACF
0.0
0.4
2
4
6
8
10
12
14
Lag

Examples



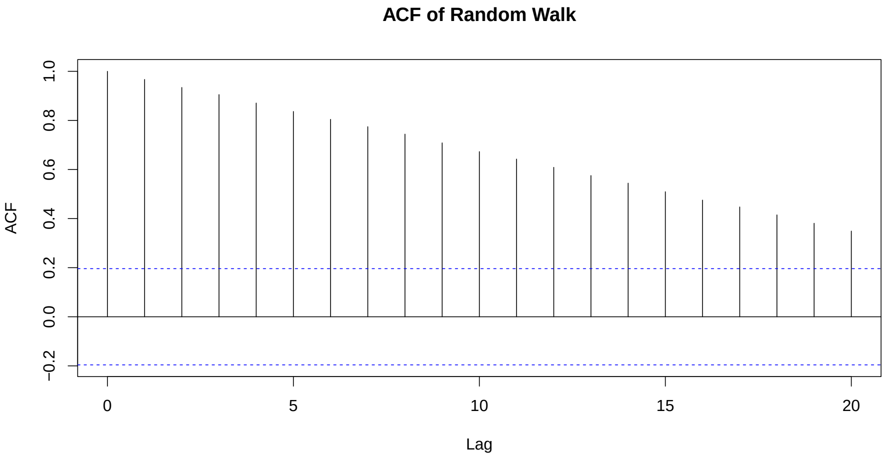
Sample ACF for ARMA(1,1), with $(\phi_1, \theta_2) = (0.8, 0.9)$
ACF
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
Lag
Sample PACF for ARMA(1,1), with $(\phi_1, \theta_2) = (0.8, 0.9)$
Partial ACF
0.8
0.6
0.4
0.2
0.0
−0.4
2
4
6
8
10
12
14
Lag

Assessing Nonstationarity

Assessing Nonstationarity through the ACF

- The ACF plot can sometimes indicate nonstationarity in a time series.
- In nonstationary series, the ACF typically decays slowly or fails to diminish quickly as the lag increases.

Example



Terminology

- Overdifferencing occurs when we choose d to be too large.
- For example, suppose that the correct model for a process $\{Y_t\}$ is an $IMA(1,1)$, that is,
$$Y_t = Y_{t-1} + e_t - \theta e_{t-1},$$
where $|\theta| < 1$ and $\{e_t\}$ is zero-mean white noise.
- The first differences are given by
$$\nabla Y_t = Y_t - Y_{t-1} = e_t - \theta e_{t-1},$$
which is a stationary and invertible $MA(1)$ process.

Terminology

- The second differences are given by

$$\begin{aligned}\nabla^2 Y_t &= \nabla Y_t - \nabla Y_{t-1} \\ &= (e_t - \theta e_{t-1}) - (e_{t-1} - \theta e_{t-2}) = e_t - (1+\theta) e_{t-1} + \theta e_{t-2} \\ &= (1 - (1+\theta) B + \theta B^2) e_t.\end{aligned}$$

- The second difference process is not invertible because

$$\Theta(x) = 1 - (1+\theta)x + \theta x^2$$

has a unit root $x = 1$.

- In this example, the correct value of d is $d = 1$. Taking $d = 2$ would be an example of overdifferencing.

Inference

- Instead of relying on the sample ACF, which may be subjective in "borderline cases", we can formally test whether or not an observed time series is stationary using the methodology proposed by Dickey and Fuller (1979).

Development

- To set our ideas, consider the model

$$Y_t = \alpha Y_{t-1} + X_t,$$

where $\{X_t\}$ is a stationary AR(k) process, that is,

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \ldots + \phi_k X_{t-k} + e_t,$$

where $\{e_t\}$ is zero mean white noise. Therefore,

$$\begin{aligned} Y_t &= \alpha Y_{t-1} + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \ldots + \phi_k X_{t-k} + e_t \\ &= \alpha Y_{t-1} + \phi_1 (Y_{t-1} - \alpha Y_{t-2}) + \phi_2 (Y_{t-2} - \alpha Y_{t-3}) \\ &\quad + \ldots + \phi_k (Y_{t-k} - \alpha Y_{t-k-1}) + e_t. \end{aligned}$$

Development

- After some algebra, we can rewrite this model for Y_t as

$$\phi^*(B)\,Y_t = e_t, \quad \text{where} \quad \phi^*(B) = \phi(B)(1 - \alpha B)$$

  and where $\phi(B) = (1 - \phi_1 B - \phi_2 B^2 - \ldots - \phi_k B^k)$ is the usual AR characteristic operator of order k.

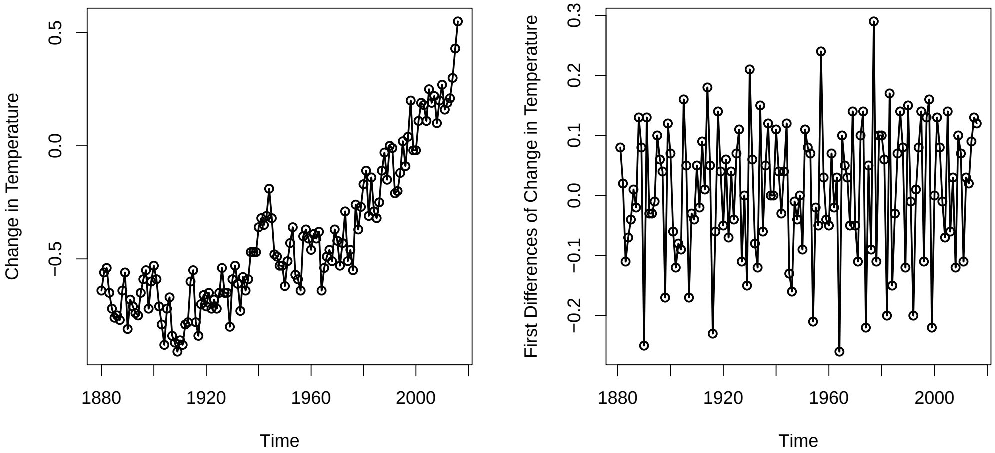
- Note that:
  - If $\alpha = 1$, then $\phi^*(B) = \phi(B)(1 - B)$, that is, $\phi^*(x)$, a polynomial of degree $k + 1$, has a unit root and $\{Y_t\}$ is not stationary.
  - If $-1 < \alpha < 1$, then $\phi^*(x)$ does not have a unit root, and $\{Y_t\}$ is a stationary $AR(k + 1)$ process.

Development

- The augmented Dickey-Fuller (ADF) unit root test therefore tests

$$H_0 : \alpha = 1 \text{ (nonstationarity)} \quad \text{vs} \quad H_1 : \alpha < 1 \text{ (stationarity)}$$

Global Temperature Data



ADF Test Output for Global Temperature Data

```
> library(tseries)
> adf.test(GT)

	Augmented Dickey-Fuller Test

data:  GT
Dickey-Fuller = -1.7516, Lag order = 5, p-value = 0.68
alternative hypothesis: stationary
```

ADF Test Output for First Difference of Global Temperature Data

```
> library(tseries)
> adf.test(diff(GT))

	Augmented Dickey-Fuller Test

data:  diff(GT)
Dickey-Fuller = -8.1457, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary
```

Discussion

When performing the ADF test, some words of caution are in order.

- When $H_0 : \alpha = 1$ is true, the AR characteristic polynomial $\phi^*(B) = \phi(B)(1 - \alpha B)$ contains a unit root. In other words, $\{Y_t\}$ is nonstationary, but $\{\nabla Y_t\}$ is stationary. This is called difference nonstationarity. The ADF procedure we have described, more precisely, tests for difference nonstationarity.

- Because of this, the ADF test outlined here may not have sufficient power to reject H_0 when the process is truly stationary. In addition, the test may reject H_0 incorrectly because a different form of nonstationarity is present (one that cannot be overcome merely by taking first differences).

Other model selection methods

Information Based Criteria: Akaike's Information Criterion (AIC)

- The Akaikes Information Criterion (AIC) says to select the $ARMA(p,q)$ model which minimizes

$$AIC = -2\ln L + 2k,$$

  where $\ln L$ is the natural logarithm of the maximized likelihood function and $k = p + q$ is the number of parameters in the model.

- The likelihood function gives the probability of the data, so we want it to be as large as possible, equivalent to wanting $-2\ln L$ to be as small as possible.

Information Based Criteria: Akaike's Information Criterion (AIC)

- The $2k$ term serves as a penalty to avoid overfitting (adhering to the Principle of Parsimony).
- AIC is an estimator of the expected Kullback-Leibler divergence, which measures the closeness of a candidate model to the truth.
- Used more generally for model selection in statistics, herein, we restrict attention to its use in selecting candidate stationary $ARMA(p, q)$ models.

Information Based Criteria: The Bayesian Information Criterion (BIC)

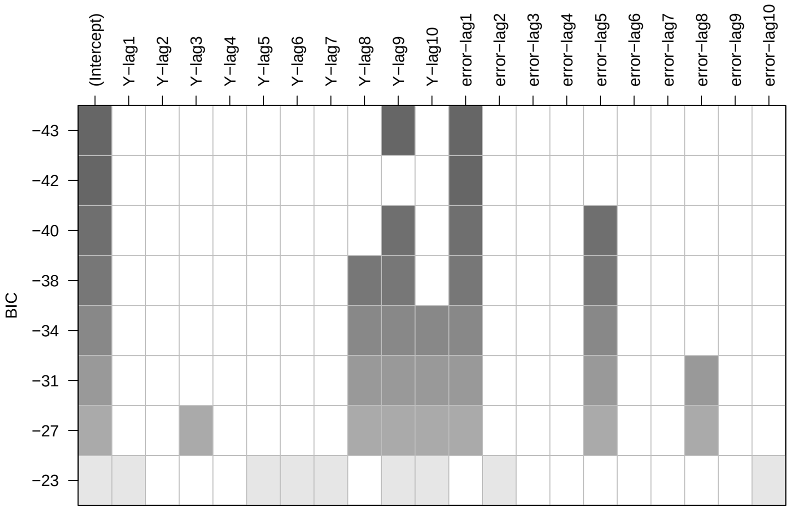
- The says to select the $ARMA(p,q)$ model which minimizes

  $$BIC = -2\ln L + k\ln n,$$

  where $\ln L$ is the natural logarithm of the maximized likelihood function and k is the number of parameters in the model and n is the sample size.

- Both AIC and BIC require the maximization of a log-likelihood function (assuming normality).

- BIC offers a stiffer penalty for overparameterized models compared to AIC since $\ln n$ will often exceed 2.




```
plot(armasubsets(diff(GT),10,10))
```

Disclaimer

- Model selection according to BIC (or AIC) does not always provide selected models that are easily interpretable. Therefore, while AIC and BIC are model selection tools, they are not the only tools available to us. The ACF and PACF may direct us to models that are different than those deemed best by the AIC/BIC.

Summary

Summary: Techniques for ARIMA Model Selection

1. Plot the data and identify an appropriate transformation if needed.
   - Understand the series for trends, seasonality, outliers, etc.
   - Check for nonconstant variance and perform suitable transformation.
2. Compute sample ACF and PACF to confirm the need for differencing.
   - Slowly decaying ACF suggests the data requires differencing.
   - Consider ADF test for stationarity and apply differencing where needed. We typically need not exceed $d = 2$ in most application.

Summary: Techniques for ARIMA Model Selection

3. Compute sample ACF and PACF to identify orders of p and q.
    - Match patterns with theoretical versions of ACF, PACF.
    - Build reasonable model with sufficient observations.

4. Use BIC and AIC for model selection consistent with data.

Remark

- Analyst may not identify a single clear-cut model. In most cases, we may identify a small number of candidate models.